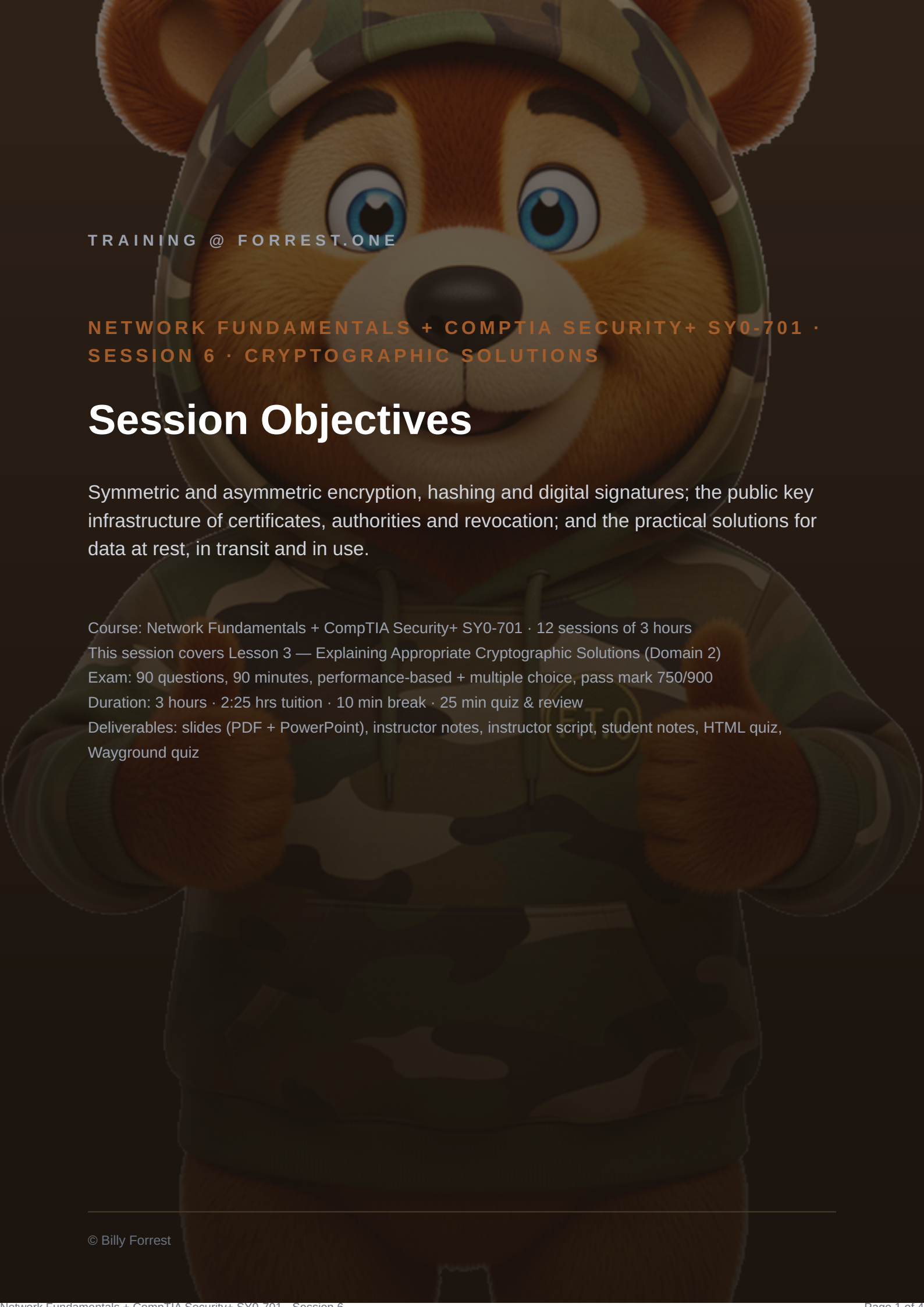




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 6 · CRYPTOGRAPHIC SOLUTIONS

Session Objectives

Symmetric and asymmetric encryption, hashing and digital signatures; the public key infrastructure of certificates, authorities and revocation; and the practical solutions for data at rest, in transit and in use.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 3 — Explaining Appropriate Cryptographic Solutions (Domain 2)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Cryptography & Public Key Infrastructure

This session covers the cryptographic mechanisms that protect data in every state, then follows them into identity and access control. It matches Lesson 3 of the Cyber Academy programme and the cryptography objectives of Domain 2 (Threats, Vulnerabilities and Mitigations), and it now carries the identity and access material that sits alongside it. No mathematics beyond key-length concepts is required: the exam tests which tool fits which job. The reference material names the standard algorithms, AES for symmetric work and RSA and ECC for asymmetric, and you should recognise each by its role rather than its mathematics.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- State the CIA triad and say which control serves confidentiality, integrity or availability, and use the terms vulnerability, threat, exploit and risk precisely.
- Explain symmetric encryption: one shared key, fast, suitable for bulk data, with a key-distribution problem; recognise AES and its 128, 192 and 256-bit key sizes.
- Explain asymmetric encryption: public and private key pairs, with the public key easy to distribute and the private key secret; recognise RSA and ECC.
- Explain key length and keyspace, and why longer keys resist brute force.
- Describe how Transport Layer Security combines an asymmetric handshake with symmetric bulk encryption.
- Describe hashing: one-way, fixed-length digests with collision resistance and the avalanche effect, used for integrity and password storage (SHA-256 or better; MD5 and SHA-1 are broken).
- Explain digital signatures: hashing plus the signer's private key, providing integrity, authentication and non-repudiation.
- Describe PKI: certificate authorities, registration authorities, X.509 digital certificates, root of trust and chains of trust.
- Explain the certificate signing request flow, subject name attributes (CN versus SAN), revocation through CRL and OCSP, and the certificate lifecycle.
- Describe key management: lifecycle, TPM and HSM cryptoprocessors, and key escrow with M-of-N control.
- Match cryptographic solutions to data states: at rest, in transit and in use, including disk, file and database encryption, perfect forward secrecy, salting and key stretching.
- Describe cryptographic applications: obfuscation and tokenisation, data masking, code obfuscation, steganography, blockchain ledgers, documentation and version control.

- Distinguish authentication from authorisation, and describe role-based access control and the identity lifecycle of provisioning, modification and deprovisioning.
- Describe the authentication factors, multi-factor authentication, one-time passwords, passwordless designs, single sign-on, SAML and federation with its risks.
- Describe central access protocols and directory services: RADIUS, TACACS+, 802.1X, LDAP and Active Directory, including LDAP ports 389, 636 and StartTLS.
- Describe access control lists and the access control models: discretionary, mandatory, role-based and attribute-based, with geofencing and time-based restrictions.

CompTIA exam context

Item	Detail
Exam	Network Fundamentals + CompTIA Security+ SY0-701
Objective area	2.8 Cryptographic solutions (SY0-701 numbering), with the identity and access objectives of Domain 4
Domain 2 weight	Threats, Vulnerabilities and Mitigations — 22%
Domain 4 weight	Security Operations and Management — 28%
Pass mark	750 on a scale of 100–900
Style	Scenario questions: which algorithm, which certificate mechanism, which data-state control, which identity technology

Lesson 3 coverage

ID	Area	What is covered
3A	Cryptographic Algorithms	Symmetric (AES), key length, asymmetric (RSA, ECC), hashing, digital signatures, TLS handshake
3B	Public Key Infrastructure	CAs, RAs, X.509, root of trust, CSR, CN and SAN, revocation, lifecycle, key management, TPM and HSM, escrow
3C	Cryptographic Solutions	Data states, disk, file and database encryption, transport encryption and PFS, salting, stretching, blockchain, obfuscation, steganography, documentation and version control
3D	Identity and Access	Authentication versus authorisation, RBAC, identity lifecycle, factors, MFA, passwordless, SSO, SAML, federation
3E	Controlling Access	RADIUS, TACACS+, 802.1X, LDAP and Active Directory, access control lists, discretionary, mandatory and attribute-based models

How this session maps to the exam

Every activity and quiz question is tagged with the exam objective it tests (e.g. 3.1 Cryptographic Algorithms). Using the session quiz to 70%+ before the exam is the fastest route to marks in this area.